Preprint

SUPPLEMENTARY MATERIAL

Supplementary Material: The TetB Field Fraction from Exchange Symmetry

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Abstract
The companion letter [1] proved that the tetrad equation produces two bounded regions related by spatial parity: TetA (the identity component) and TetB (the parity component). This note derives the fraction of the total gravitational field response carried by each sector. The exchange symmetry between TetA and TetB, combined with rational bijectivity on the projective line, uniquely determines the TetA fraction as $\mu(x) = x/(x+1)$ and the TetB fraction as $1-\mu(x) = 1/(x+1)$, where $x = g/a_0$ is the dimensionless acceleration. The derivation uses no field equation, no coupling polynomial, and no equilibrium postulates. It requires only the parity relation between the two bounded regions of the tetrad equation together with the rational-bijection classification on the projective line.

The two sectors

The companion letter [1] proves that the spatial sector of the tetrad equation $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ produces exactly two bounded tetrahedral regions. Under spatial parity $P : e^i \to -e^i$, the identity component gives TetA and the parity component gives TetB. Their union is the stellated octahedron. The tetrad equation is quadratic in the frame field: TetA and TetB are its two roots in the parity direction, and the metric $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ is the quadratic form common to both.

Because TetA and TetB together constitute the two bounded regions of the tetrad equation, the gravitational field response at each point partitions between them. Let $\mu(x)$ denote the fraction of the total field response carried by TetA at dimensionless acceleration $x = g/a_0$, so that $1-\mu(x)$ is the fraction carried by TetB. The two fractions sum to unity because μ measures the relative partition, not the absolute field strength.

The exchange symmetry

Spatial parity exchanges TetA and TetB, swapping the two fractions: if μ is the TetA fraction before the exchange, then $1-\mu$ is the TetA fraction after it.

The natural variable for a two-sector partition is the odds ratio between TetA and TetB:

$$\Omega = \frac{\mu}{1-\mu} = \frac{\text{TetA fraction}}{\text{TetB fraction}}. \tag{1}$$

Under exchange, the numerator and denominator swap, so $\Omega \to 1/\Omega$ by construction. This inversion is algebraic, requiring no assumption about how the acceleration transforms.

Two endpoint conditions follow from the partition definitions:

- $\mu(0) = 0$: at zero acceleration, the field sits entirely in TetB. Equivalently, $\Omega(0) = 0$.
- $\mu(x) \to 1$ as $x \to \infty$: at large acceleration, the field sits entirely in TetA. Equivalently, $\Omega \to \infty$.

The dimensionless acceleration $x = g/a_0$ is constructed from the metric, which is determined by the 40 symmetric (Christoffel) components of the connection; these are the TetA sector [1]. The variable x therefore measures TetA-sector dominance: $x > 1$ where TetA dominates, $x < 1$ where TetB dominates, and $x = 1$ at the parity midpoint. Since x is intrinsically a TetA-sector quantity, exchange inverts it to $1/x$; combined with $\Omega \to 1/\Omega$, the full exchange symmetry requires $\Omega(1/x) = 1/\Omega(x)$.

Classification

The map $\Omega(x)$ must be rational, to preserve the algebraic structure of the tetrad equation, and bijective, so that a single acceleration determines a unique partition. Rationality is inherited from the polynomial character of the tetrad equation itself; bijectivity is the physical requirement that the field be uniquely decodable from the observed acceleration. A rational bijection on the projective line is necessarily fractional linear [2]:

$$\Omega(x) = \frac{ax+b}{cx+d}. \tag{2}$$

The unique solution

Proposition. *Let $\Omega(x) = (ax+b)/(cx+d)$ be a rational bijection satisfying $\Omega(1/x) = 1/\Omega(x)$ and $\Omega(0) = 0$. Then $\Omega(x) = x$.*

Proof. The endpoint condition $\Omega(0) = b/d = 0$ requires $b = 0$. The map reduces to $\Omega(x) = ax/(cx + d)$.

Substituting into $\Omega(1/x) = 1/\Omega(x)$ and cross-multiplying gives $a^2x = (c + dx)(cx + d)$. Comparing coefficients of x^2, x^1, and x^0:

$$x^2 : \quad cd = 0, \tag{3}$$

$$x^1 : \quad a^2 = c^2 + d^2, \tag{4}$$

$$x^0 : \quad cd = 0. \tag{5}$$

If $d = 0$: $\Omega = a/c$, a constant, which is not bijective. Rejected.

If $c = 0$: $a^2 = d^2$, so $a = \pm d$. The branch $a = -d$ gives $\Omega(x) = -x$, which is incompatible with $\Omega > 0$ for $x > 0$. Therefore $a = d$ and

$$\Omega(x) = x. \tag{6}$$

□

The TetA and TetB fractions

From $\Omega(x) = x$, the TetA fraction follows by normalisation:

$$\mu(x) = \frac{\Omega}{1 + \Omega} = \frac{x}{x + 1}. \tag{7}$$

The TetB fraction is the complement:

$$\boxed{1 - \mu(x) = \frac{1}{x + 1}.} \tag{8}$$

This is the fraction of the total gravitational field response carried by TetB, the parity component of the tetrad equation, at dimensionless acceleration x.

At large acceleration ($x \gg 1$), the TetB fraction vanishes: the field response is carried almost entirely by TetA. At the parity midpoint ($x = 1$), TetA and TetB contribute equally: $\mu = 1/2$, $1 - \mu = 1/2$. At low acceleration ($x \ll 1$), the TetB fraction approaches unity: the field response is carried almost entirely by TetB.

Consequences for the connection

The companion letter [1] establishes that TetA is the identity component and TetB is the parity component under spatial parity $P : e^i \to -e^i$. The affine connection $\Gamma^I{}_{JK}$ decomposes under index exchange ($J \leftrightarrow K$) into 40 symmetric components (Christoffel) and 24 antisymmetric components (torsion).

A torsion-free connection is uniquely determined by the metric and its first derivatives. The metric $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ is bilinear in the frame field; under $P : e^i \to -e^i$, both factors acquire a sign flip and the two signs cancel, leaving the metric invariant under TetA $\leftrightarrow$ TetB exchange. A torsion-free connection, built entirely from this invariant metric, therefore cannot distinguish TetA from TetB. This root-blindness is a structural property of the quadratic map $\phi : e \mapsto \eta(e, e)$: the image of a 2-to-1 map does not resolve which pre-image produced it. TetA and TetB are geometrically distinct regions [1]; their indistinguishability under the torsion-free connection reflects the character of ϕ, not a physical equivalence of the two roots. The torsion sector is the necessary carrier of the TetA/TetB distinction at the connection level.

General relativity's zero-torsion condition $\Gamma^I{}_{[JK]} = 0$ removes this carrier from the connection. In the partition language of this note, this sets $1 - \mu = 0$ and therefore $\mu = 1$: the full field response is attributed to a single effective sector.

The exchange symmetry proves that $\mu \neq 1$. The TetB fraction is $1/(x + 1)$, not zero. The field response that general relativity removes from the connection is:

$$\Delta\mu(x) = \frac{1}{x + 1}. \tag{9}$$

The dark-to-baryon ratio

If the TetB field response is interpreted as a mass contribution (as the cold dark matter framework does), the ratio of the removed response to the retained response is:

$$\frac{\Delta\mu}{\mu} = \frac{1/(x + 1)}{x/(x + 1)} = \frac{1}{x}. \tag{10}$$

This is an algebraic identity of the constitutive law. It does not depend on any galaxy model, halo profile, or observational data.

The cold dark matter framework decomposes the observed gravitational field as $g_{\text{obs}} = g_{\text{baryon}} + g_{\text{DM}}$. By definition, $g_{\text{baryon}} = \mu \cdot g_{\text{obs}}$ and $g_{\text{DM}} = (1 - \mu) \cdot g_{\text{obs}}$. Therefore:

$$\frac{g_{\text{DM}}}{g_{\text{baryon}}} = \frac{1 - \mu}{\mu} = \frac{1}{x}. \tag{11}$$

The dark-to-baryon mass ratio that the cold dark matter framework fits with a multi-parameter halo profile reduces to $1/x$: the reciprocal of the dimensionless acceleration. This follows from the exchange symmetry of the two parity-related sectors of the tetrad equation, with no free parameters and no halo model.

Summary

The tetrad equation produces two bounded regions related by spatial parity: TetA (identity) and TetB (parity). The exchange symmetry, together with rational bijectivity, uniquely determines the partition of the

gravitational field response between them:

$$\text{TetA:} \quad \mu(x) = \frac{x}{x+1}, \tag{12}$$

$$\text{TetB:} \quad 1 - \mu(x) = \frac{1}{x+1}. \tag{13}$$

The torsion sector of the affine connection is the necessary carrier of the TetA/TetB distinction: a torsion-free connection, built from the parity-invariant metric alone, cannot distinguish the two chambers. General relativity's zero-torsion condition removes this carrier. The removed fraction is $\Delta\mu = 1/(x+1)$, and the resulting dark-to-baryon ratio is $1/x$. The torsion-sector field fraction and the cold-dark-matter residual are algebraically identical under the exchange-symmetric partition.

References

[1] S. Nelson, *The Coordinate-Space Geometry of the Tetrad Frame*, Class. Quantum Grav. (2026), submitted (CQG-115428).

[2] R. Hartshorne, *Algebraic Geometry* (Springer, New York, 1977).